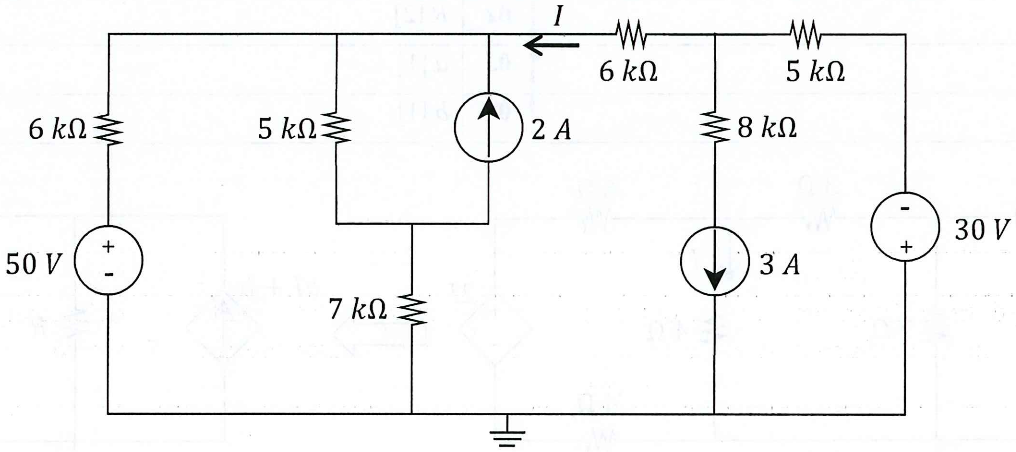
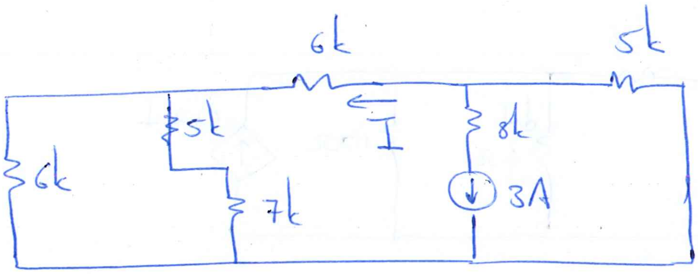
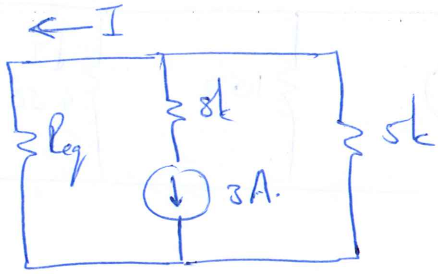
EBN111 2017 CT2 ⇒ Memo

| Question 1 [2] | | Vraag 1 [2] | |
|---|---|---|---|
| 01 | Use superposition to find the exclusive contribution of the 3A current source to the current *I*. | 01 | Maak gebruik van superposisie en bepaal die bydra van die 3 A stroombron tot die stroom *I* alleenlik. |




$$R_{eq} = 6k + 6k \| (5+7)k$$

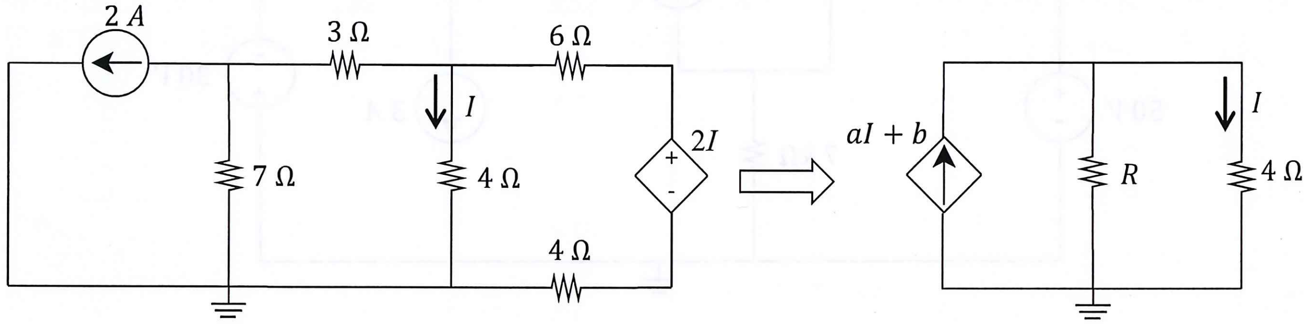
$$= 6k + \left(\frac{6 \times 12}{6+12}\right)k$$

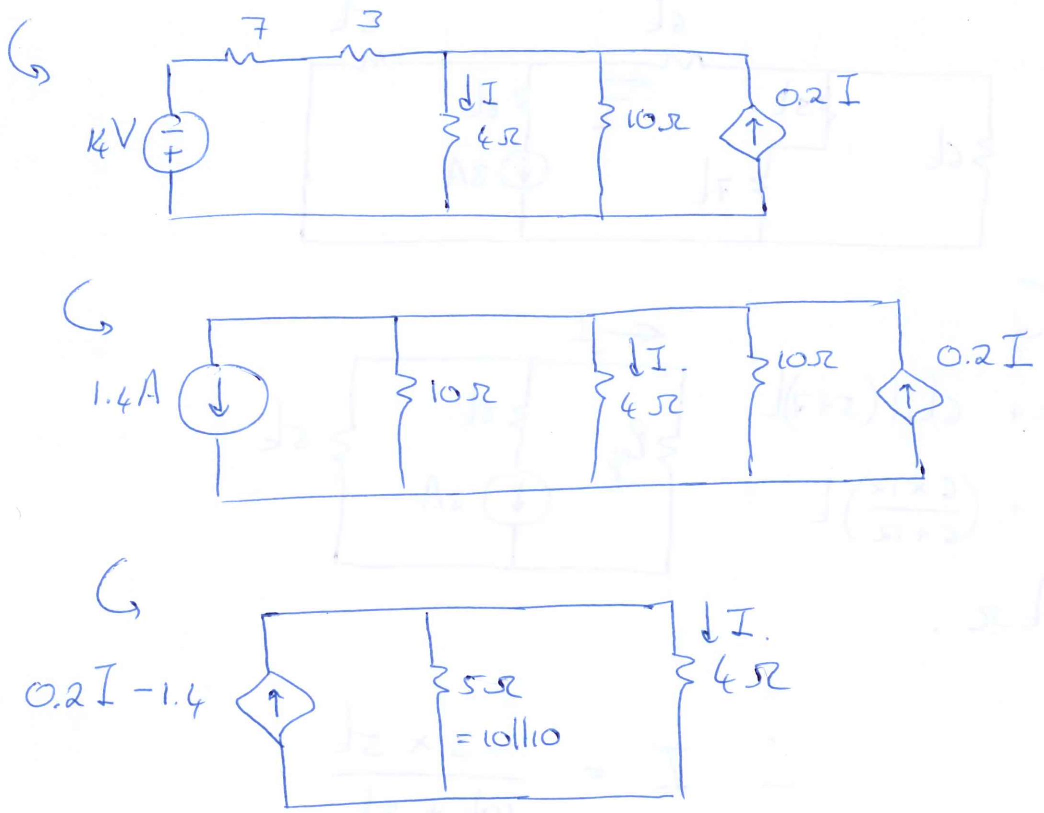
$$= 10k\Omega.$$

$$\therefore I = \frac{-3 \times 5k}{10k + 5k}$$

$$= \underline{\underline{-1\,A}}$$

| Question 2 - 4 [4] | | Vraag 2 - 4 [4] | |
|---|---|---|---|
| Simplify the circuit on the left by means of source transformation so that it looks like the one on the right. Determine: | | Vereenvoudig die stroombaan aan die linkerkant met behulp van brontransformasie sodat dit lyk soos die kringbaan aan die regterkant. Bepaal: | |
| **02** | *R* [2] | **02** | *R* [2] |
| **03** | *a* [1] | **03** | *a* [1] |
| **04** | *b* [1] | **04** | *b* [1] |



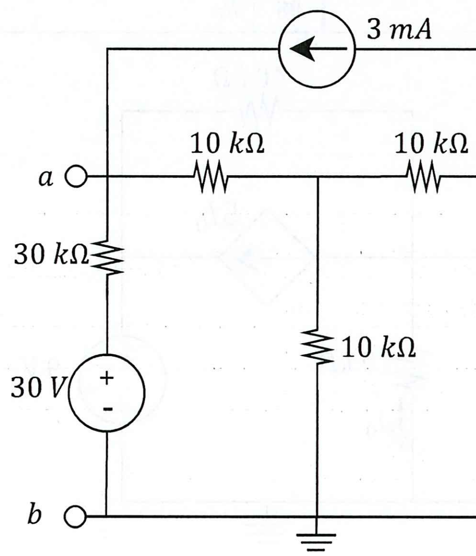




$R = 5\Omega$

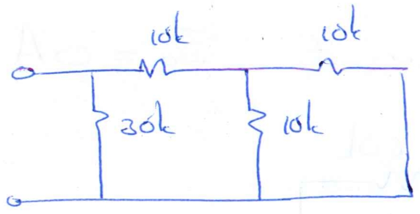
$a = 0.2$

$b = -1.4.$

| Question 5 & 6 [4] | | Vraag 5 & 6 [4] | |
|---|---|---|---|
| Determine the Thevenin equivalent voltage $V_{TH}$ and resistance $R_{TH}$ w.r.t. terminals *a*-*b*. | | Bepaal die Thevenin ekwivalente spanning $V_{TH}$ en weerstand $R_{TH}$ met verwysing tot terminale *a* en *b*. | |
| **05** | $V_{th}$ | **05** | $V_{th}$ |
| **06** | $R_{th}$ | **06** | $R_{th}$ |




$\underline{R_{TH}}$

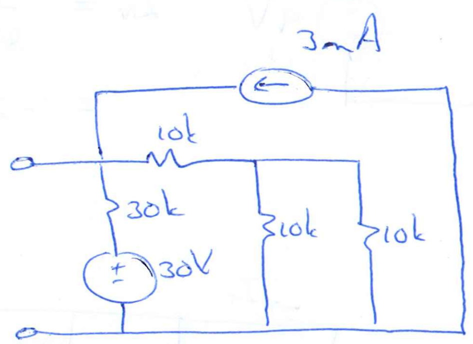



$$R_{eq} = (10 + 10 \| 10) \| 30$$

$$= (10 + 5) \| 30$$

$$= \frac{15 \times 30}{15 + 30} = \underline{10\,k\Omega}$$

$\underline{V_{TH}}$

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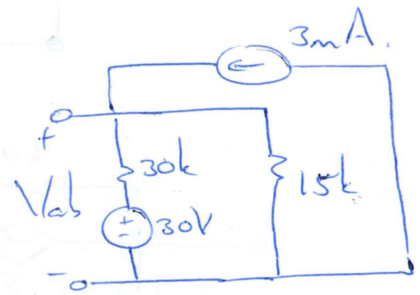


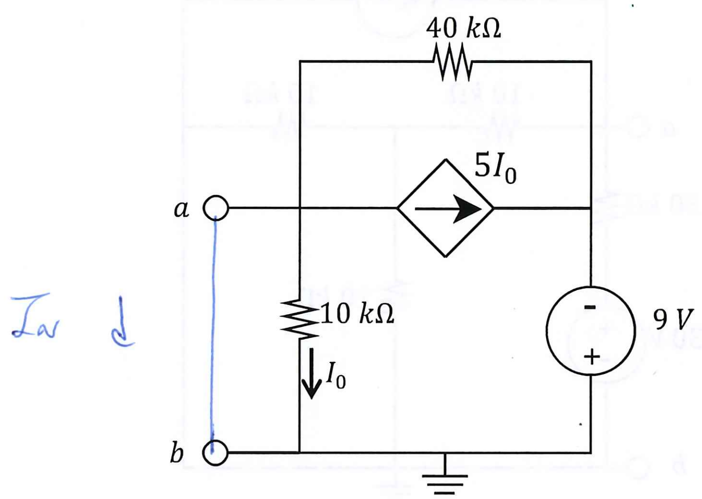

$$\frac{V_{ab} - 30}{30k} + \frac{V_{ab}}{15k} = 3mA$$

$(\times 30)$ $\quad V_{ab} - 30 + 2V_{ab} = 90$

$$3V_{ab} = 120$$

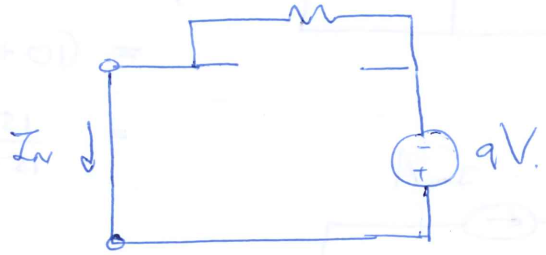
$$V_{ab} = \underline{60V}$$

| Question 7 & 8 [4] | | Vraag 7 & 8 [4] | |
|---|---|---|---|
| Determine the Norton equivalent current $I_N$ and resistance $R_N$ w.r.t. terminals *a*-*b*. | | Bepaal die Norton ekwivalente stroom $I_N$ en weerstand $R_N$ met verwysing tot terminale *a* en *b*. | |
| 07 | $I_N$ | 07 | $I_N$ |
| 08 | $R_N$ | 08 | $R_N$ |



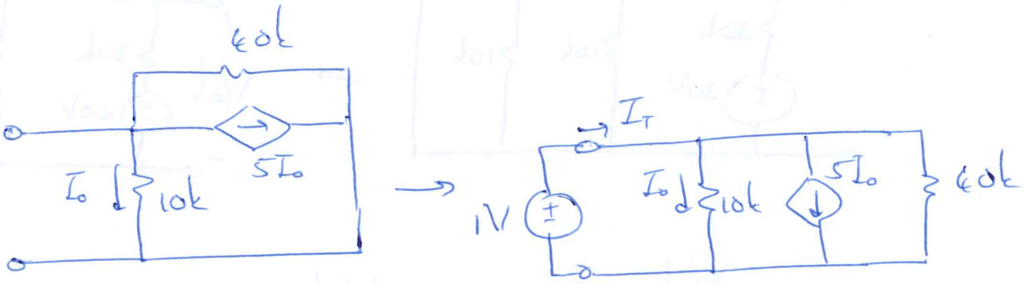

$I_N$

* Because of short circuit: $I_o = 0A$.
* Therefore:




$$I_N = \frac{-9}{40k} = \underline{-225\mu A}$$

$R_N$

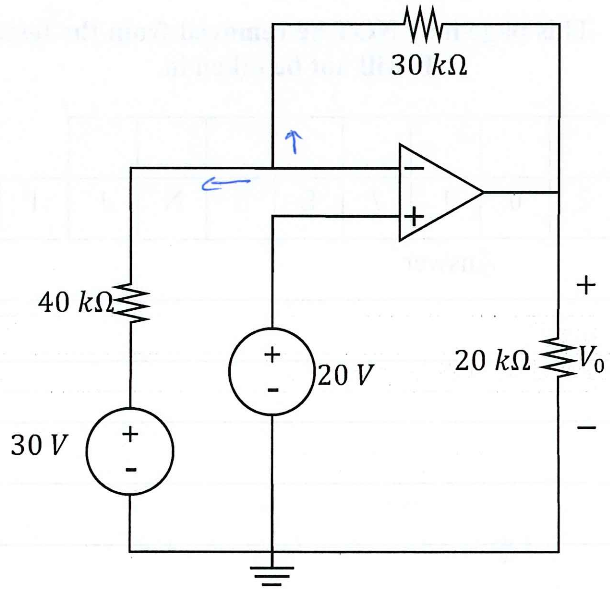



$$I_T = \frac{1}{10k} + \frac{1}{40k} + 5\left(\frac{1}{10k}\right)$$

$$= \frac{6}{10k} + \frac{1}{40k} = \frac{24+1}{40k} = \frac{25}{40k}.$$

$$R_T = \frac{1}{I_T} = \frac{40k}{25} = \underline{1.6 k\Omega}$$

| Question 9 [2] | | Vraag 9 [2] | |
|---|---|---|---|
| 09 | Find $V_0$ for the circuit below. | 09 | Bepaal $V_0$ in onderstaande kringbaan. |




* $V_+ = 20V$ $\therefore$ $V_- = 20V$

* KCL at $V_-$

$$\frac{20-30}{40k} + \frac{20-V_o}{30k} = 0$$

$$30k\left(\frac{-10}{40k}\right) = V_o - 20$$

$$V_o = 20 - \frac{30}{4}$$

$$= 20 - 7.5$$

$$= \underline{12.5V}$$

Memo

**PRACTICE ANSWER SHEET /**
**OEFEN ANTWOORDBLAD**

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| Assessment ID | 2 | 0 | 1 | 7 | E | B | N | 1 | 1 | 1 | C | 2 | |

| | Answer | Unit |
|---|---|---|
| 01 | $I =$ $-1$ | $A$ |
| 02 | $R =$ $5$ | $\Omega$ |
| 03 | $a =$ $0.2$ | |
| 04 | $b =$ $-1.4$ | |
| 05 | $V_{th} =$ $40$ | $V$ |
| 06 | $R_{th} =$ $10$ | $k\Omega$ |
| 07 | $I_N =$ $-225$ | $\mu A$ |
| 08 | $R_N =$ $1.6$ | $k\Omega$ |
| 09 | $V_0 =$ $12.5$ | $V$ |